

The graph-topology Fredholm problem has a full K-theoretic answer

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Status: literature_already_answered. Let $\mathcal{CF}^{\text{sa}}$ be the graph-topology space of closed, dense-domain, self-adjoint Fredholm operators on an infinite-dimensional separable complex Hilbert space. Prokhorova's Theorem C identifies this space, by natural homotopy equivalences, with classical models for odd K -theory. Hence

$$\mathcal{CF}^{\text{sa}} \text{ classifies } K^1, \quad \pi_1(\mathcal{CF}^{\text{sa}}) \cong \mathbb{Z}.$$

The spectral-flow map on π_1 is therefore an isomorphism, up to sign.

The original problem

Lesch considers the set $\mathcal{CF}^{\text{sa}}$ of unbounded self-adjoint Fredholm operators with the graph metric d_G . The source paper establishes that this space is path connected and that spectral flow gives a surjective homomorphism

$$\text{sf} : \pi_1(\mathcal{CF}^{\text{sa}}, d_G) \longrightarrow \mathbb{Z}.$$

It then asks explicitly: determine this fundamental group, and decide whether the space is a classifying space for the K^1 functor [1, p. 3].

The homotopy-equivalence theorem

Write $\mathcal{R}_F^{\text{sa}}(H)$ for the space of regular self-adjoint Fredholm operators equipped with the graph topology. This is the same operator space and topology as in Lesch's question. Prokhorova places it in a commutative diagram with the nontrivial component of bounded self-adjoint Fredholm operators, its Riesz-topology counterpart, and the standard Fredholm-unitary model. Her Theorem C states that every map in this diagram is a homotopy equivalence and concludes that every space in the diagram classifies K^1 [2, pp. 5–6, Theorem C].

In particular, the theorem answers the stronger half of Lesch's problem affirmatively:

$$(\mathcal{CF}^{\text{sa}}, d_G) \simeq U_F,$$

where U_F is a classical representing space for odd complex K -theory. The paper also explicitly notes that the graph-topology space is path connected [2, p. 5].

Fundamental group and spectral flow

Since the space represents odd complex K -theory, Bott periodicity gives

$$\pi_1(\mathcal{CF}^{\text{sa}}, d_G) \cong \pi_1(U_F) \cong K^1(S^1) \cong \mathbb{Z}.$$

Lesch had already proved that spectral flow is surjective onto $\mathbb{Z}$. After identifying the domain with $\mathbb{Z}$, this surjective homomorphism is multiplication by $+1$ or -1 . Thus spectral flow is an isomorphism; the sign depends only on the normalization convention.

The result is not new to this run. Indeed, Prokhorova’s introduction records that Joachim proved the K^1 analogue for regular self-adjoint operators in 2003, and presents her theorem as a transparent proof with explicit natural homotopy equivalences [2, pp. 2–3]. The modern theorem directly settles both clauses of the source problem and also makes the homotopy model fully explicit.

Scope and verification

The exact problem statement was checked in the official arXiv source and in the included local compilation. The answering statement was checked in the official PDF of arXiv:2110.14359v3, especially Theorem C and its introductory setup. The run’s registry, solution, attempt, and proof-gap indexes contained no prior resolution of this exact target.

References

- [1] Matthias Lesch, *The uniqueness of the spectral flow on spaces of unbounded self-adjoint Fredholm operators*, arXiv:math/0401411; in *Spectral Geometry of Manifolds with Boundary and Decomposition of Manifolds*, Contemp. Math. **366** (2005), 193–224.
- [2] Marina Prokhorova, *Spaces of unbounded Fredholm operators. I. Homotopy equivalences*, arXiv:2110.14359v3, 2025.